

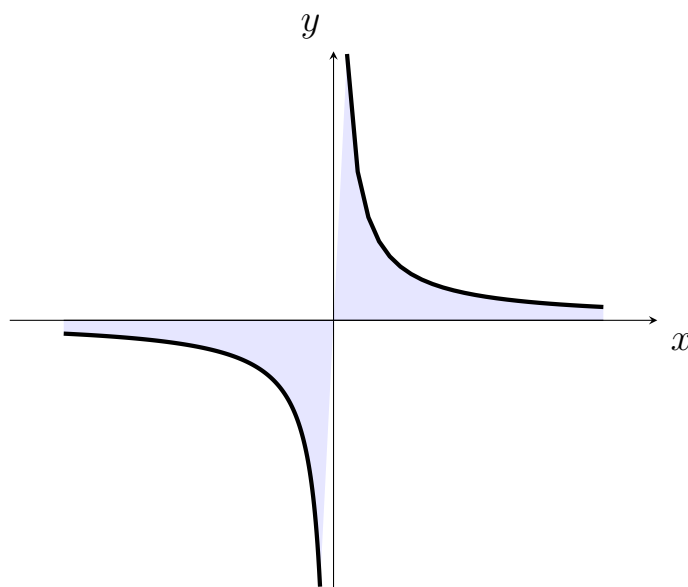
7.8a Improper Integrals

Before this section we have required two things of our definite integrals:

1. The domain of integration, $[a, b]$, is _____.
2. The range of the integrand is _____ on the domain.

We say that the integrals described above are _____ integrals.

How might we find the area under the curve $f(x) = \frac{1}{x}$ on $(-\infty, \infty)$?



Infinite Limits of Integration

Definition

Improper integrals of Type I are definite integrals with _____ limits. Suppose f is a function.

1. If f is continuous on $[a, \infty)$, then

$$\int_a^{\infty} f(x) dx = \quad .$$

2. If f is continuous on $(-\infty, b]$, then

$$\int_{-\infty}^b f(x) dx = \quad .$$

3. If f is continuous on $(-\infty, \infty)$, then

$$\int_{-\infty}^{\infty} f(x) dx =$$

where c is any real number.

Note:

1. If the limit is finite, then we say that the definite integral _____ and the limit is the _____ of the improper integral.
2. If the limit is not finite, we say the definite integral _____.

Example 1

Find the area under the curve $y = \frac{\ln(x)}{x^2}$ for $x \geq 1$.

Example 2

For what values of p does the integral $\int_1^\infty \frac{dx}{x^p}$ converge?
What is the value when it converges?

Example 3

Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$.

Integrands with Vertical Asymptotes

Definition

Improper integrals of Type II are definite integrals with integrands that are _____ within the limits of integration.

Suppose f is a function.

1. If f is continuous on $(a, b]$, but not continuous at a , then

$$\int_a^b f(x) dx = \quad .$$

2. If f is continuous on $[a, b)$, but not continuous at b , then

$$\int_a^b f(x) dx = \quad .$$

3. If f is continuous on $[a, c) \cup (c, b]$, but not continuous at c in (a, b) , then

$$\int_a^b f(x) dx =$$

If the limit is finite, then we say that the definite integral _____ and the integral is the _____ of the limit. If the limit is not finite, we say the definite integral _____.

What does condition 3 tell us?

Example 4

Does $\int_0^{\pi/2} \tan(\theta) d\theta$ converge? If so, what is its value?

Example 5

Evaluate $\int_0^3 \frac{1}{(s-1)^{2/3}} ds$.